Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/16/2024
PI Name:	Stirling, Leia Ph.D.		
Project Title:	HCAAM VNSCOR: Responsive Multimodal Human-Automation Communication for Augmenting Human Situation Awareness in Nominal and Off-Nominal Scenarios (80NSSC20K0409)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109	**Congressional District:**	12
Comments:	NOTE: PI moved to University of Michigan in fall 2019; previous affiliation was Massachusetts Institute of Technology		
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	12/04/2019	**End Date:**	12/03/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/03/2024 per NSSC information (Ed., 11/10/23).		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Blossom, Jonathon (NASA Jet Propulsion Laboratory) Atkins, Ella Ph.D. (Virginia Tech) Liu, Andrew Ph.D. (Massachusetts Institute of Technology)		
Grant/Contract No.:	80NSSC20K0409		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note April 2020: Continuation of "HCAAM VNSCOR: Responsive Multimodal Human-Automation Communication for Augmenting Human Situation Awareness in Nominal and Off-Nominal Scenarios," grant 80NSSC19K0703, with the same Principal Investigator (PI) Leia Stirling, Ph.D., due to PI move to University of Michigan from Massachusetts Institute of Technology in fall 2019] This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR). Crew extravehicular activity (EVA) is limited on spaceflight missions. Multiple, small robotic spacecraft with varying levels of autonomy are needed to perform tasks that might have been completed by an astronaut (e.g., an exterior surface inspection or repair). Crews on long duration exploration missions (LDEM) will have less access to ground support during task operations. As a result, they will need to process more information and communicate with autonomous robots effectively to ensure tasks are progressing safely and on schedule. The objective of these studies is to investigate the use of augmented reality (AR) multimodal interface displays and communication pathways for improving human-robot communication, situation awareness (SA), trust, and task performance. This will lead to developing guidelines for designing human-robot system interactions that enable operational performance for crews on spaceflight missions. The specific aims are to: 1) Develop a simulation testbed for examining communication between human-robot teams. 2) Develop a hardware testbed for examining communication between human-robot teams. 3) Evaluate human SA, trust, and task performance within a short duration and long-duration ground-based study (simulation and/or hardware) through testing various interface communication modalities and information displays. 4) (Option) Perform additional studies for alternate parameters of interest that could be tested using the study testbeds. Additional parameters include timing and persistence of information, gesture command mapping, varying the levels of robot automation, evaluating precision enabled by each command mode.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Augmented Reality (AR) has the opportunity to support decision making across a variety of use-case scenarios, including but not limited to manufacturing, automated vehicles, military training, and entertainment. This research compares AR to other visual modalities for telerobotics applications, specifically considering robotic control and anomaly inspection. Results from this study can inform how AR is integrated for task-specific applications, as there may be tasks that have increased benefit from AR, whereas others may have additional considerations that emerge.
Task Progress:	The objective of these studies is to investigate the use of augmented reality (AR) multimodal interface displays and communication pathways for improving human-robot communication, situation awareness (SA), trust, and task performance. This will lead to developing guidelines for designing human-robot system interactions that enable operational performance for crews on spaceflight missions. During this reporting period, our team performed synergistic follow-on studies using the simulation space station environment developed in a previous reporting period. The first study examined how the motion plan trajectory shape influenced a decision to takeover control during a docking maneuver from an egocentric viewpoint. Here takeovers occurred when the human supervisor deemed that the agent was going to an incorrect docking location, or a collision could occur. The second study examined how viewpoint (egocentric vs. exocentric) influenced the decision to takeover control during a docking maneuver. We observed that both path trajectory and viewpoints influenced the decisions to takeover control. We also observed that there were effects on the decision to handover control (i.e., when the human supervisor determined the agent was performing correctly and they no longer needed to monitor performance). Findings from these studies will be summarized for the NASA HIDH. During the reporting period, our team also completed analysis from studies to support gaps in the NASA Human Integration Design Handbook (HIDH) for AR interfaces. We compared performance on sensorimotor and neurovestibular assessment tasks using physical objects and within AR. The sensorimotor task was a multi-directional Fitts' Law target acquisition task, while the neurovestibular tasks included three clinical balance assessments (Four Square Step test (FSST), the Star Excursion, tandem walking/TW), and three operational tasks (capsule ingress and egress, geology sampling, and obstacle avoidance). For the sensorimotor task, the touchscreen modality yielded improved performance over AR as measured by accuracy, precision, error rates, throughput, and movement time. AR designers can improve performance when designing AR interfaces by selecting larger buttons when accuracy and efficiency are required and by embedding perception cues to button target surfaces such as depth and proximity cues. For the neurovestibular tasks, while participants were able to perform all assessment tasks successfully, their strategies differed when comparing AR performance to physical object use. Task completion times were longer when administered in AR. In the Ingress/Egress task and the Obstacle Avoidance task, head pitch increased while overall head motion and torso motion decreased. These adaptations align with maintaining head stability for viewing the holographic content. The participants were able to complete all tasks within AR and meaningful measures of task performance and postural control were obtained, indicating that AR may be a useful instrumentation solution with embedded sensors to evaluate task performance and postural control. However, care should be taken when comparing performance within AR to other assessment modalities. Results from these studies support research gaps identified in NASA's Human Research Roadmaps, provide design guidance for AR in NASA's Human Integration Design Handbook, and support determining whether AR is a viable tool for evaluating astronauts' vestibular performance throughout mission timelines.
Bibliography Type:	Description: (Last Updated: 11/25/2025)

Abstracts for Journals and Proceedings	Apodaca B, Atkins E, Stirling L. "Effect of orbital dynamics on fuel use in sampling-based plans for proximity operations." 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, October 1-5, 2023. 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, October 1-5, 2023. , Oct-2023
Articles in Peer-reviewed Journals	Weiss H, Tang J, Williams C, Stirling L. "Performance on a target acquisition task differs between augmented reality and touch screen displays." Applied Ergonomics. 2024 Apr;116:104185. https://doi.org/10.1016/j.apergo.2023.104185 ; PubMed PMID: 38043456 , Apr-2024
Articles in Peer-reviewed Journals	Allred AR, Weiss H, Clark TK, Stirling L. "An augmented reality hand-eye sensorimotor impairment assessment for spaceflight operations." Aerosp Med Hum Perform. 2024 Feb 1;95(2):69-78. https://doi.org/10.3357/AMHP.6313.2024 ; PubMed PMID: 38263106 , Feb-2024
Papers from Meeting Proceedings	Larson H, Stirling L. "Autonomous Spacecraft Motion Plan Characteristics Influence Perceived Path Appropriateness." Human Factors and Ergonomics Society Annual Meeting, Phoenix, AR, September 9-13, 2024. Human Factors and Ergonomics Society Annual Meeting, Phoenix, AR, September 9-13, 2024. , Sep-2024
Papers from Meeting Proceedings	Apodaca B, Atkins E, Stirling L. "RRTZ: A Path Planner for Zero Gravity." IEEE Aerospace Conference, Big Sky, MT, March 2-9, 2024. IEEE Aerospace Conference, Big Sky, MT, March 2-9, 2024. , Mar-2024